The University of Zambia

in association with

ZCAS University

BFS322 TREASURY MANAGEMENT

2nd DECEMBER 2019

12:30 – 15:30 HOURS

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer Three (3) questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

Monetary Policy Committee Statement

Bank of Zambia Policy Rate Adjusted Upwards by 125 basis points to 11.50%

The Monetary Policy Committee (MPC) at its November 18-19, 2019 meeting decided to raise the Policy Rate by 125 basis points to 11.50%. Inflation is now projected to remain above the upper bound of the 6-8% target range over the entire forecast horizon, with some of the upside risks identified in our previous MPC Statement materialising. The decision to raise the Policy Rate is therefore intended to counter inflationary pressures that include exchange rate pass-through effects and bring inflation back to the target range in the medium-term and hence support overall macroeconomic stability. The Committee recognises that to address the prevailing economic challenges, monetary policy actions alone are not sufficient. They need to be complemented by the implementation of corrective measures by fiscal authorities and other key public policy makers. Moreover, implementation of measures that address high fiscal deficits, debt levels and debt service, as well as liquidity constraints and dismantling domestic arrears remain critical to maintaining overall macroeconomic stability and attaining sustainable economic growth.

(Extract from the November 2019, BOZ MPC statement.)

(i) The Bank of Zambia (BOZ) considers a number of economic factors when deciding on whether to adjust the policy rate upwards or downwards. Identify and describe at least three (3) of these factors. **(10 marks)**

(ii) The Financial Markets Department is in charge of the implementation of monetary policy at the Bank of Zambia. Critically analyse the role of this department in the implementation of monetary policy. **(20 marks)**

(iii) Discuss the Ministry of Finance treasury functions. **(10 marks)**

(Total = 40 marks)

SECTION B: **Attempt any THREE questions in this section**

Question 2

Nkongole Enterprises Limited is a manufacturer of exotic furniture. You have been recently appointed as the company Treasurer. The company has been facing some working capital challenges recently, which has led it to rely on its overdraft facility with the bank on which it pays 20% p.a. in interest.

One of your first tasks as a Treasurer is to examine the management accounts of the company and you have been given the following information:

	2018	**2017**	**2016**
Cash	K0	K0	K5,000
Debtor days	90	60	30
Creditor days	120	100	45
Stock turnover per month	2	2.5	4

Required

i. Based on the above figures alone, advise management on the causes of the working capital problems that the company is facing. **(5 marks)**

ii. Explain why it is important for a firm to manage its working capital. **(5 marks)**

iii. One of the ways of managing working capital is to manage how much credit the company grants to its customers. Suggest ways in which the company can assess the creditworthiness of its customers. **(10 marks)**

(Total = 20 marks)

Question 3

The Treasurer of Mwango Enterprises Limited has forecast that the firm will have surplus cash available for investment in the foreseeable future. It is the treasurer's role to ensure that these surplus funds are invested in appropriate investment instruments.

(i) Outline the factors that the Treasurer will have to take into consideration when selecting the appropriate investment vehicle. **(6 marks)**

(ii) List and discuss any five (5) investment vehicles available in Zambia that the Treasurer can investment the surplus funds in. **(10 marks)**

(iii) Suggest any two (2) investment strategies that the Treasurer can use when investing the surplus funds. **(4 marks)**

(Total = 20 marks)

Question 4

Herstatt risk is derived from the German Herstatt Bank which was closed in 1974 for insolvency. The operations of the bank were suspended in the middle of a foreign exchange transaction. Banks that had transferred currency to Herstatt Bank never received the agreed upon exchange currency. Therefore the term Herstatt risk now refers to the risk of the transactional parties not settling the currency exchange.

(i) Describe two (2) types of settlement and two (2) types of transaction in the clearing and settlement process. **(4 marks)**

(ii) The Continuous Link Settlement (CLS) system was established in response to the settlement risk discussed above. Briefly discuss the operation of this system and how it helps in reducing settlement risk. **(4 marks)**

(iii) The Bank of Zambia has also put in place measures to try and improve the payments are settlement systems in Zambia. Write short notes on the following payment and settlement systems and for each system suggest any two advantages of the system.

(a) Zambia Interbank Payment and Settlement System (ZIPSS) also known as the Real-time Gross Settlement (RTGS) System. **(4 marks)**

(b) Electronic Funds Transfer (EFT). **(4 marks)**

(c) The Cheque Image Clearing (CIC) system. **(4 marks)**

(Total = 20 marks)

Question 5

The treasury department requires information which is not normally available through a company's standard accounting system. Consequently, treasury systems are needed so as to integrate information from a variety of sources which the treasurer can use to efficiently perform their tasks.

(i) Suggest the three (3) priorities that a treasury system should address and under each category discuss any two (2) specific functions technology needs of the treasury system. **(6 marks)**

(ii) The treasury system must be able to improve efficiency and have adequate controls. Discuss the efficiency and control issues that a good treasury system should be able to address. **(10 marks)**

(iii) Treasury systems require a significant initial capital outlay i.e. they cost a lot of money. Provide two possible arguments that you can use to convince management of a company to spend on the treasury system. **(4 marks)**

(Total = 20 marks)

END OF EXAMINATION

Marking Key

Question 1

(i) Factors that BOZ takes into consideration when adjusting the policy rate:

Inflation

- The main target for adjusting the policy rate is inflation.
- Inflation is the general increase in the price of goods and services over a period of time.
- BOZ has a target range of inflation which at the moment is 6-8 percent.
- The policy rate will be adjusted in order to maintain inflation in this target range.

Exchange Rate

- The rate of exchange between the kwacha and other major currencies, especially the US dollar is another consideration.
- BOZ would want to see a stable exchange rate of the kwacha against the major currencies.
- An unstable currency can have a negative effect on economic growth and inflation

Economic Growth

- BOZ considers the effect of changes in the policy rate on economic growth
- Economic growth as measured by GDP will be affected by the level of interest rates
- When the policy rate is adjusted upwards, it signals an upward adjustment to lending rates, which impacts on the cost of borrowing for firms

Financial stability

- The health of the financial institutions is also a factor for BOZ when deciding to adjust the policy rate.
- Unstable exchange and interest rates can have a negative effect on financial institutions, especially the banking sector.

(ii) Role of the Financial Markets Department in implementation of monetary policy:

- Implements monetary policy and act as a fiscal agent of the government
- Plays critical role in contributing to BOZ objective of achieving price stability

Implement monetary policy

- On 2nd April 2002 BOZ introduced the Policy Rate as the key interest rate to signal monetary policy stance
- MPC meets every quarter to discuss policy rate
- Policy Rate currently at 11.50%

- Adjustment to rate guided by economic indicators such as output, expected inflation and exchange rate
- Bank has established corridor that defines the band within which overnight interbank rate is allowed to fluctuate
- Bank's stance on whether to buy or sell financial assets determined by level of short term interest rates particularly in overnight market
- Aim is to steer overnight interbank rate towards BOZ Policy Rate
- Achieved by department undertaking OMO to either supply or withdraw liquidity using various financial instruments
- A stable liquidity position is maintained in financial system in line with desired level of interest rates
- Influencing short term interest rates at appropriate levels assists in achieving price and financial system stability
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Acting as fiscal agent of the government;

- Executed through issuance and redemption of GRZ securities
- Conducts primary auctions of GRZ securities, redemptions and interest payments as well as maintaining GRZ securities databases
- Advises and provides domestic debt projections and estimations of domestic debt service to MOF for annual national budget purposes

Administering the Overnight Lending Facility;

- A liquidity financing facility which commercial banks can access in need of liquidity from BOZ at day end
- OLF complements other refinancing facilities such as TB Rediscount Facility and Intraday Credit Facility
- Primary purpose of OLF is to enhance operations of money markets by assuring commercial banks of liquidity assistance from BOZ

Managing foreign exchange reserves;

- Assets denominated in foreign currency held by monetary authorities
- Consist of cash, financial instruments, gold and SDRs
- International reserves held for a number of reasons:

Undertaking foreign exchange dealings;

- Department undertakes foreign exchange dealings that enables Bank to respond to core function of price and financial system stability
- Zambia has flexible exchange rate system
- Exchange rate policy management aimed at ensuring relative stability in exchange rate
- Executed by interventions in FX market

(iii) Ministry of Finance treasury functions:

- MOF derives mandate from part XVL of the constitution of Zambia
- Also guided by Statutory and portfolio functions as stipulated in GRZ Gazette Notice No. 76 of 2016
- Charged with economic national development planning, budgeting and financial management responsibilities
- Ministry that deals with Country's treasury function
  The Ministry of Finance must manage government financial assets, propose economic and financial policy, and coordinate and supervise these actions as empowered by law.
- Prepare the annual fiscal budget and issue adequate regulations for its execution.
- Manage government financial assets.
- Propose bills related to the management of government employees, particularly bills related to staffing, salaries, benefits and pensions.
- Amortize debt and coordinate financial activities carried out within the scope of its power, for the different ministries and their related entities.
- Be aware of and report on any initiative of a financial nature involving public expenditure and indebtedness before
- Participate in the negotiation of international free trade and financial agreements.
- Exercise all powers and rights as conferred on it in the Law.

Question 2

(i) The working capital problems could be caused by the following: (1) for each point

- The company has run down its cash over the years
- This could be due to the fact that the cash conversion cycle has been lengthening over the years
- Its customers are taking longer to pay, as revealed by the worsening debtor days. This has the effect of tying up cash in the receivables
- It is taking the company longer to turn over its stock. This means that cash is increasingly tied up in unsold stock
- As a result of the worsening cash position of the company, it finds itself unable to meet its commitments as revealed by the worsening creditor days position.

(ii) The efficient management of working capital is important from the point of view of both liquidity and profitability. Poor management of working capital means that funds are unnecessarily tied up in idle assets, hence reducing liquidity and also reducing the ability to invest in productive assets such as plant and machinery, thereby affecting profitability.

(iii) Sources of information for credit standing of customers: (2 marks each)

- Bank references- most banks in Zambia offer references on their customers
- Trade references- suppliers already giving credit to the customer can give useful information about how good a customer is on baying bills
- Published information – customer's annual accounts and reports will give some idea of general financial position of company and its liquidity
- Credit agencies- agencies such as the Credit Reference Bureau in Zambia keeps information of individual and company borrowings from different banks
- Company's own sales record- for an existing customer, the sales ledger will show how prompt a payer the company is

Question 3

(i) When considering various forms of cash investment, the treasurer should first consider the following:

- Safety of the principal being invested
- Maturity and marketability
- Yield

(ii) Investment options:

a. Bankers acceptances

- Banks sometimes guarantee (or accept) corporate debt, usually when they issue a loan to a corporate customer, and then sell the debt to investors.
- Because of the bank guarantee, they are viewed as obligations of the bank

b. Bonds near maturity dates

- A corporate bond may not mature for many years, but one can purchase a bond that is close to its maturity date.
- Their tends to be a minimal risk of loss (or gain) on the principal amount of this investment, since there is low risk that interest rates will change so much in the short period of time left to maturity date that will impact its value

c. Certificate of deposits

- Are term bank deposits, typically having durations of up to two years.
- Pay a fixed interest rate, though some variable rate CDs are available
- Perception that they are more secure than CP since they are issued by banks

d. Commercial Paper

- Large corporations issue short-term notes that carry higher yields than government debt issues
- There is an active secondary market for them, so they are very liquid
- CP is generally not secured, however investing in CP issued by 'blue chip' companies reduces default risk
- Most CP matures in 30 days or less, and rarely matures in greater than 270 days, in order to avoid registration requirements of the SEC
- CP is issued at a discount

e. Money market fund

- Package of government instruments, usually composed of treasury bills, notes, and bonds assembled by a fund management company
- Investment is highly liquid with many investors putting in funds for as little as a day
- There are varying levels of risk between different money market funds, since some funds are more active in trying to outperform the market (with an attendant increase in risk

f. Repurchase agreement (REPO)

- A package of securities that an investor buys from a financial institution, with an agreement that the institution will buy it back at a specified date
- Most commonly used for the overnight investment of excess cash
- Interest rate is equal to or less than money market rate

g. Treasury bills issuance

- Issued by the Central Bank on behalf of government to finance budget deficits
- Considered as risk free investments
- Issued at a discount to their face value
- Have lower yields than most other forms of investments

(iii) Investment Strategies:

i. Earnings credit strategy

- At the most minimal level where the treasurer does nothing and leaves idle balances in the corporate bank accounts
- Earnings from idle balances offset its service fees
- Ideal for firms with minimal cash balances

ii. Matching strategy

- Treasurer matches the maturity date of an investment to the cash flow availability dates listed on the cash forecast.
- For example, ABC Co. cash forecast indicates that ZMW80,000 will be available for investment immediately, but must be used in two months for a capital project
- The treasurer can invest the funds in a two-month instrument, such that its maturity date is just prior to when the funds will be needed
- This investment strategy is more concerned with short-term liquidity than return on investment

iii. Laddering Strategy

- Involves creating a set of investments that have a series of consecutive maturity dates
- For example, ABC Co. cash forecast indicates that ZMW150,000 of excess cash will be available for the foreseeable future, and its investment policy forbids any investments having a duration of greater than three months
- The treasurer could invest the entire amount in a three month instrument to take advantage of the higher interest rate
- However there is a risk that some portion of the cash will be needed sooner
- Instead the treasurer will break the available cash into thirds, and invests ZMW50,000 in a month instrument, another in a two month instrument, and the final ZMW50,000 in a three month instrument
- As each investment matures, the treasurer reinvests it into a three month instrument
- By doing so, ABC always has ZMW50,000 of the invested amount coming due within one month or less.
- This improves liquidity, while still taking advantage of longer term interest rates

iv. Tranched cash flow strategy

- Requires the treasurer to determine what cash is available for short, medium, and long term investment
- He can then adopt different investment criteria for each investment tranche e.g.:
- Short term tranche is treated as cash that may be needed for operational requirements on a moments notice. Return on investment is not a key criterion-instead the treasurer focuses on very high levels of liquidity
- Medium term tranche- includes cash that may be required for use within the next 3 to 12 months and usually for predictable events such as periodic tax or dividend payments. Given the much higher level of predictability in this tranche, the treasurer can accept longer term maturities with moderate levels of volatility that have higher returns on investment
- The long term tranche includes cash for which there is no planned operational use, and which the treasurer feels can be safely invested for at least one year. The priority for this tranche shifts more in favor of a higher return on investment, with an attendant potential for higher levels of volatility and perhaps short term capital loss, with a reduction in the level of liquidity

v. Riding the yield curve

- A strategy of buying longer term securities and selling them prior to their maturity dates
- Strategy works when interest rates on short term securities are lower than rates on longer term securities (i.e. normal yield curve)
- For example ABC Co. has ZMW75,000 of cash available for investment for a three month period
- The treasurer invests in a six month security and sells it after three months, achieving a higher than usual rate of return
- However, if the yield curve had changed during the interim, so that short term rates were higher than long term rates, then the treasurer would have sold the security and earned a below market return on the investment

vi. Credit rating strategy

- The treasurer buys the debt of a company that may be on the verge of having its credit rating upgraded

- By doing so, the investment ends up earning a higher interest rate than would other investments with comparable credit rating.
- This strategy works best when the treasurer is very familiar with the debt issuer and has confidence in the credit assessment
- However, the company's stated investment policy may prevent the treasurer from buying debt below a fairly high credit rating
- Also it is difficult to time a possible credit rating upgrade to be within the term of an investment
- Finally investing into lower grade debt increases the risk of an outright default on payments by the debtor

Question 4

(i) Settlement types

Gross basis

- where each bank pays the total amount owed.
- Payments handled through a gross basis are more likely to have a requirement for immediate execution, where payment instructions are processed separately for each individual transaction

Net basis

- where a large number of transactions are accumulated and offset against each other, with only the net difference being transferred between banks
- Payments usually wait until the end of the day, when all transactions between the banks are summarized and offset against each other

Transaction types:

High-value payments

- must be executed immediately
- Because of the time constraints, high value payments cannot wait for end-of-day netting and so are settled on a gross basis, with an immediate transfer between the banks

Low-value payments

- does not require immediate execution and tends to be for smaller amounts.
- These payments are handled through a net settlement system

(ii) Continuous Link Settlement (CLS)

- This is called "settlement risk" or "Herstatt risk" after the German Bank that failed.
- To avoid this settlement risk, while also speeding up the settlement process, the CLS system was created by a number of banks
- This system is operated by CLS Bank International, of which the founding banks are shareholders
- Other banks can submit their foreign exchange transactions through these member banks.
- Only a select number of foreign currencies are transacted in the CLS system

(iii) Payment systems in Zambia:

(a) Real Time Gross Settlement System (RTGS)

- ZIPSS is an electronic payment system in which processing of transactions for settlement takes place continuously on a transaction basis in real time
- ZIPSS was developed to improve the management of payment risks related to high value payments.
- It provides a more secure platform for making both large value payments
- It also provides quicker final and irrevocable payments

(b) **Electronic Funds Transfer (EFT)**

- The Electronic Funds Transfer (EFT) system is a simple, safe and speedy way to make and collect payments by electronic transfer

- Safe and secure transfer of funds from one account to another
- Payment of wages, salaries and pensions
- Payment of utility bills
- Disbursement of loans and collection interest payments by financial institutions

(c) The Cheque Image Clearing System (CIC)

- The Cheque Image Clearing (CIC) system is a cheque processing and clearing system and allows customers belonging to one bank to write cheques belonging customers of another bank.
- Similar to Electronic Funds Transfer (EFT) system, the CIC system is operated by the Zambia Electronic Clearing House Ltd (ZECHL).
- With the implementation of the truncation system in 2014, substantial improvement in turn-around time in cheque processing have been achievement.
- Credit to customer accounts is given on a T+1 basis.

Question 5

(i) Treasury Technology needs:

(a) Cash position: treasurer overriding obligation is to ensure that the company has adequate cash to fund operations. The following technology needs are critical:

- Cashbook balance tracking
- Multibank reporting of balance information
- Cash pooling management

(b) Foreign exchange transactions: some companies have little foreign business that FX transactions are negligible. Larger firms with established treasury operations will likely need ongoing purchase and sale of multiple currencies. The following technology needs arise:

- Rate feeds from Bloomberg or Reuters
- Intercompany netting capability

(c) Hedging: for those companies that choose to hedge their interest rates or foreign exchange rate positions, the following technology requirements maybe needed

- Rate feeds from Bloomberg or Reuters
- Exposure modeling capability
- Hedge deal-making capability
- Hedge confirmation processing

(ii) Efficiency and control issues:

(a) Efficiency Issues

Minimise data entry- system should never require manual entry of a transaction into the system more than once and ideally should involve automated data collection and posting

Work flow processing- if supervisory approval is required, the system should electronically route transaction to correct supervisor for approval

(b) Control Issues

Audit trail- all treasury transactions should result in a clearly defined audit trail that identifies who made a transaction and the date, amount and accounts impacted by the transaction

Segregation of duties- the system should limit access to certain modules and require approval of key transactions

Warning indicators- the system should automatically notify users if transaction confirmations have not been received, if hedging policies are being violated, if there are negative cash balances etc

(iii) Treasury management systems is costly

- The expense can be justified by the fact that it automates many rote treasury task

- The firm can justify the cost of acquiring the system by looking at the cost of a failed manual system, such as a spread sheet error
- Another option is to enlist the internal auditors, who can argue that the risk of a transactional error may result in a Sarbanes-Oxley control breach that would have to be revealed in a company's filings